

HISHAAM ALI ABBASI

hishaamabbasi@gmail.com • 07710 812020 • hishaamabbasi.co.uk • [LinkedIn](#) • [GitHub](#)

PROFILE

Final year BEng Robotics Engineering student (Predicted First, IMechE accredited) with end-to-end experience across embedded firmware, mechanical design, control systems, SLAM, and DevOps infrastructure. Proven ability to take systems from concept to working prototype. Full case studies at hishaamabbasi.co.uk. Seeking a graduate role in robotics, embedded systems, or product development.

EDUCATION

BEng Robotics Engineering, Queen Mary University of London — Predicted First | IMechE Accredited *2022–2026*
Key modules: Numerical Methods & Data Science (86%), Control Systems (78.9%), Modelling & Control of Robotic Systems (75.2%), Intro to AI (74.7%), Instrumentation & Measurements (76.1%).

Sutton Grammar School — A Levels: Maths (A), Physics (A), Computer Science (B); 12 GCSEs incl. Maths (9), CS (8), Physics (8)

EXPERIENCE

Engineering Design & Research Intern, Alvis Robotics — EOD Systems, Birmingham *Aug 2025*

- Diagnosed and resolved a firmware fault on a proprietary sensor/camera module integrated onto a legacy Northrop Grumman EOD UGV; traced to incompatible driver boards and restored full operational functionality.
- Resolved a mud-seized active pitch mechanism on a tracked EOD platform via full disassembly, fault diagnosis, and clearance; returned the unit to operational condition.
- Supported R&D planning with the Managing Director on next-generation EOD robot development.

PROJECTS

Fracture Immobilisation Trainer — Final Year Dissertation | ESP32, Silicone, Haptics, FDM *2025–2026*

- Led anatomical core workstream on a 5-person team; designed a 1:1 forearm chassis (NASA 50th-percentile anthropometric data) with internal channels for fluidics, sensors, and haptic actuators.
- Selected PETG-CF via cantilever deflection study across 4 FDM polymers on elastic stability grounds (flat apparent Young's modulus, no plastic yielding vs 15% drift in PLA).
- Built dual-durometer silicone soft-tissue system (Dragon Skin 30A muscle, Ecoflex 00-30 skin) with Hall-effect alignment sensing via ESP32; functional prototype at <£500 vs £2,000 commercial comparator.

TurtleBot3 SLAM & Autonomous Navigation — ROS, Gmapping, Python, Raspberry Pi *EMS627U*

- Implemented differential drive kinematics and Gmapping SLAM fusing LiDAR scans with wheel odometry into a real-time 2D occupancy grid on a physical TurtleBot3.
- Developed a Python obstacle-avoidance node using LiDAR thresholding; migrated the full stack from Gazebo simulation to physical hardware and resolved ROS/Ubuntu networking blockers.

Formula Student Suspension Redesign, Queen Mary Racing — Ansys FEA, CAD, Vehicle Dynamics *Oct 2023–Present*

- Redesigned 8 critical suspension components (wishbones, uprights, bell crank) for the new car platform using Ansys FEA; achieved 12% unsprung mass reduction while maintaining peak stiffness, contributing to a targeted 0.3 s lap-time improvement.

Home Server & Self-Hosted Infrastructure — Proxmox, Docker, Cloudflare, AWS *2025–Present*

- Built a Proxmox VE hypervisor with Nvidia GPU passthrough to LXC for NVENC hardware transcoding; resolved kernel pinning, cgroup device passthrough, and driver conflicts.
- Deployed Jellyfin, Nextcloud (8 TB), Duplicati with encrypted nightly backups to AWS S3 Glacier; configured Cloudflare Tunnel and Zero Trust across 5 domains, eliminating open ports.

Additional projects: 2-axis gyro-stabilised camera gimbal (C++, PID); 3D-printed sim racing pedals (load-cell braking, PETG-CF); Ender 3 V3 SE Klipper upgrade; BoxTurtle multi-material filament system; robotic arm; web development.

LEADERSHIP

- **Explorer Scout Leader, Scouting Movement (UK)** — *2012–Present*. Chief Scout Diamond Award (highest level in Scouting); founded a new Explorer section from inception; plan and deliver weekly sessions, expeditions, and residential camps for 18+ young people.
- **Flight Sergeant, Combined Cadet Force** — *Sep 2018–Jul 2023*. Progressed to the highest NCO rank; taught the CCF syllabus, led teams on residential camps, and mentored younger cadets across varied environments.

TECHNICAL SKILLS

Programming: Python (Advanced), Java (Advanced), C++ (Intermediate), C, SQL, HTML/CSS/JavaScript

Robotics & Control: ROS/ROS2, SLAM (Gmapping), PID, Kinematics, Gazebo, Sensor Fusion, ESP32, Arduino, Raspberry Pi

CAD & Manufacturing: Onshape (Advanced), SolidWorks (Int.), Creo (Int.), Fusion 360, Ansys FEA, FDM 3D Printing (Advanced), Klipper, Orca Slicer

Infrastructure: Linux (Debian/Ubuntu), Proxmox VE, Docker/Compose, LXC, Nginx, Cloudflare Tunnel/Zero Trust, AWS S3/Glacier, Git

Other: Full UK driving licence | Native English | Conversational Urdu